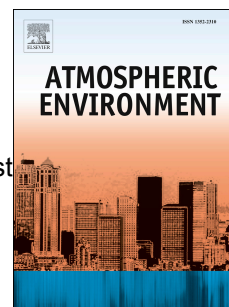


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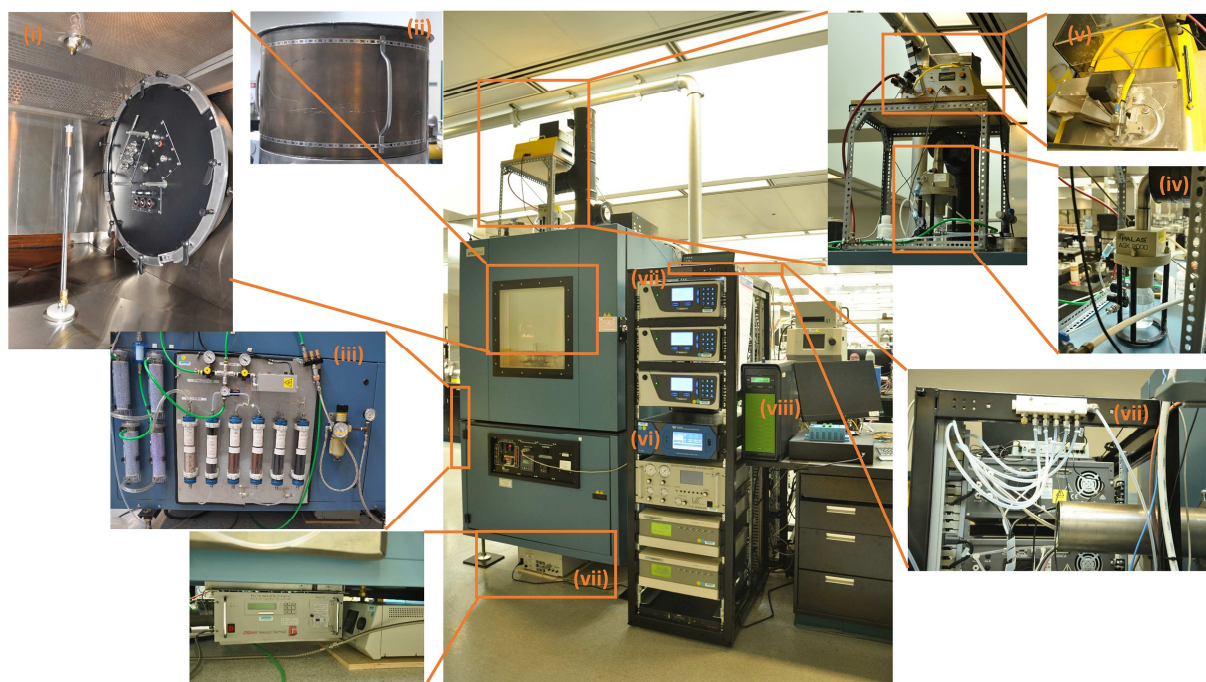
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Development of an environmental chamber for evaluating the performance of low-cost air quality sensors under controlled conditions

Vasileios Papapostolou, Hang Zhang, Brandon J. Feenstra, Andrea Polidori*

South Coast Air Quality Management District, Diamond Bar, CA 91765, USA

5 Correspondence to: Andrea Polidori (apolidori@aqmd.gov)

Highlights

- A state-of-the-art laboratory chamber was developed to evaluate air quality sensors.
- This chamber generates stable and reproducible environmental conditions.
- Sensors should be tested under a wide range of T and RH conditions.
- 10 • A rigorous sensor testing method was also developed.
- As technology improves a more standardized testing protocol should be developed.

Abstract

A state-of-the-art integrated chamber system has been developed for evaluating the performance of low-cost air quality sensors. The system contains two professional grade chamber enclosures. A 1.3 m³ stainless-steel outer chamber and a 0.11 m³ Teflon-coated stainless-steel inner chamber are used to create controlled aerosol and gaseous atmospheres, respectively. Both chambers are temperature and relative humidity controlled with capability to generate a wide range of environmental conditions. The system is equipped with an integrated zero-air system, an ozone and two aerosol generation systems, a dynamic dilution calibrator, certified gas cylinders, an array of Federal Reference Method (FRM), Federal Equivalent Method (FEM), and Best Available Technology (BAT) reference instruments and an automated control and sequencing software. Our experiments have demonstrated that the chamber system is capable of generating stable and reproducible aerosol and gas concentrations at low, medium, and high levels. This paper discusses the development of the chamber system along with the methods used to quantitatively evaluate sensor performance. Considering that a significant number of academic and research institutions, government agencies, public and private institutions, and individuals are becoming interested in developing and using low-cost air quality sensors, it is important to standardize the procedures used to evaluate their performance. The information discussed herein provides a roadmap for entities who are interested in characterizing air quality sensors in a rigorous, systematic and reproducible manner.

Keywords

Low-cost sensors, sensor evaluation, environmental chamber, testing protocol, particle pollutants, gaseous pollutants

1 Introduction

30 Studies have shown that air pollutants, such as fine particulate matter (PM_{2.5}), ozone (O₃), volatile organic compounds (VOC), and nitric oxides (NO_x), can cause serious respiratory diseases, cardiovascular disorders, and other adverse health effects (Heck

et al. 2013; Weichenthal et al. 2011). Conventionally, air quality and pollutant concentrations are monitored by the federal government as well as state and local regulatory agencies using sophisticated and expensive fixed-site instruments (Snyder et al. 2013). The number of monitoring sites is thus limited by the cost of instrumentation and availability of trained personnel to operate and maintain such equipment. Because of their relatively low spatial density, available fixed air monitoring sites are mostly designed to characterize air quality over a wide geographical area. However, they do not typically provide the granularity that is often necessary to fully understand local air quality conditions. Due to recent technological advancements in micro-sensors, embedded systems, and wireless networks, manufacturers have begun marketing low-cost and relatively easy-to-use air quality sensors. These devices, provided they produce reliable data, can significantly augment and improve current ambient air monitoring capabilities. A wide range of sensor applications is now changing the paradigm of air pollution monitoring (Snyder et al. 2013). In its 2014 – 2018 Strategic Plan, the United States Environmental Protection Agency (U.S. EPA) has recognized the need to extend the existing air pollution monitoring to lower cost measurements (EPA 2014). Herein, a sensor is considered low-cost if its market cost is less than \$2000. If a device is presented as a multi-pollutant sensor device, then the cost per pollutant type should be less than \$2000. With the recent commercialization of low-cost and easy-to-use devices, susceptible groups and individuals such as children, seniors, asthmatics, pregnant women, and people interested in measuring air pollution in their communities can monitor air quality and assess potential personal exposure. Citizen scientists and community groups have now access to a wealth of information to better understand how air pollution may impact their neighborhoods (Deville Cavellin et al. 2016; Jiao et al. 2015). Air quality sensors deployed near industrial facilities, such as those for fence-line monitoring applications, can provide empirical data to supplement existing ambient air monitoring infrastructure (Pikelnaya et al. 2013). Portable sensor mounted on a mobile vehicle can also be used to monitor urban air quality and map the spatial variation of traffic related emissions (Hagler et al. 2010; Van den Bossche et al. 2015).

The development of reliable low-cost air quality sensors is complex. These devices have to detect one or more specific target pollutants while being inert to interferent species and meteorological parameters. They also have to be calibrated to give accurate readings, which often requires the need of specialized and expensive reference instrument. Furthermore, various algorithms and processing procedures are used to convert the sensor's signal to air pollution concentrations. Therefore, not all low-cost sensors are reliable or able to provide meaningful air quality information (Williams et al. 2014). Consequently, their performance needs to be fully characterized under various pollutant levels and different environmental conditions to assure data quality.

Environmental chambers have been an indispensable tool in studying gas-phase atmospheric chemistry (Cocker et al. 2001) and pollutant exposure (Papapostolou et al. 2013) because they can provide controlled environments. Over the past years, there have been several studies on sensor performance evaluation that involved environmental chambers. At EPA, a glass exposure chamber was constructed to evaluate the performance of O₃, nitrogen dioxide (NO₂) and VOC sensors (Williams et al. 2014; Williams et al. 2015). Temperature and relative humidity (RH) were controlled by an air conditioning system, supplemented with heating pads, dry ice, and a water bubbler. Important parameters such as linear correlation coefficient, detection limit, concentration resolution, response time, and temperature and RH influences were examined. Yet, due to the chamber size limitations and restricted resources, the intra-model variability, the effect of interferents and weather conditions were not tested in those studies. Joint Research Center (JRC) has published a series of technical reports and papers describing its efforts in evaluating and calibrating gaseous sensors (Spinelle et al. 2014). In the JRC laboratory testing approach, an "O"-shaped ring-tube exposure chamber was developed to evaluate an ozone sensor (model B4, Alphasense, UK) under controlled temperature, RH, wind velocity, and gaseous interferent concentrations. The sensor reported highly linear output, good precision, and little baseline drift, but it was sensitive to NO₂ and was affected by hysteresis due to RH variations. Although this work was conducted for an ozone sensor, it provided important guidelines for other gas sensors evaluation. In another study, three particle sensors, including

Shinyei PPD42NS, Samyoung DSM501A and the Sharp GP2Y1010AU0F, were evaluated in a customized acrylic chamber where particulate atmosphere was created by burning incense (Wang et al. 2015). The method was limited in scope, as the system could not generate stable and reproducible particulate environment, thus was not appropriate for systematic evaluation of PM sensors of different types. Additionally, a Shinyei PPD42NS particulate sensor was also evaluated in a chamber, but only under ambient temperature and RH conditions (Austin et al. 2015). Similar studies (Northcross et al. 2013; Sousan et al. 2016a; Sousan et al. 2016b) have definitely expanded our understanding on the potential and limitations of low-cost air monitoring devices. Nonetheless, there has not been any effort to develop methods, protocols, and procedures to systematically evaluate the performance of low-cost particle and gaseous sensors under a wide range of environmental conditions.

Herein, we describe the development of a state-of-the-art chamber system for the performance evaluation of low-cost air quality sensors. To the best of our knowledge, this integrated chamber system is the first that can generate stable and reproducible environmental conditions with diverse temperature, RH, and known PM and gas concentration profiles. The chamber system is coupled with an array of FRM, FEM, and BAT reference instruments for comparison purposes. In this paper, we focus on the development of methods and the validation of the chamber's ability to generate a wide range of environmental conditions. Indicative laboratory experiments are presented to exemplify the practical application of the chamber system and the testing methods. A summary of all available laboratory testing results conducted within Air Quality – Sensor Performance Evaluation Center (AQ-SPEC) can be found on a dedicated website (www.aqmd.gov/aq-spec). Our sensor evaluation results indicate that sensor performance can be better characterized when parameters such as accuracy, precision, detection limit, climate susceptibility, and the effect of interferences are investigated systematically.

2 Methodology

2.1 Chamber system overview

A chamber system, designed by the AQ-SPEC team, hardware developed and integrated by AmbiLabs (Warren, RI), has been installed inside the South Coast Air Quality Management District's Chemistry Laboratory (Figure 1 and Figure S-1).

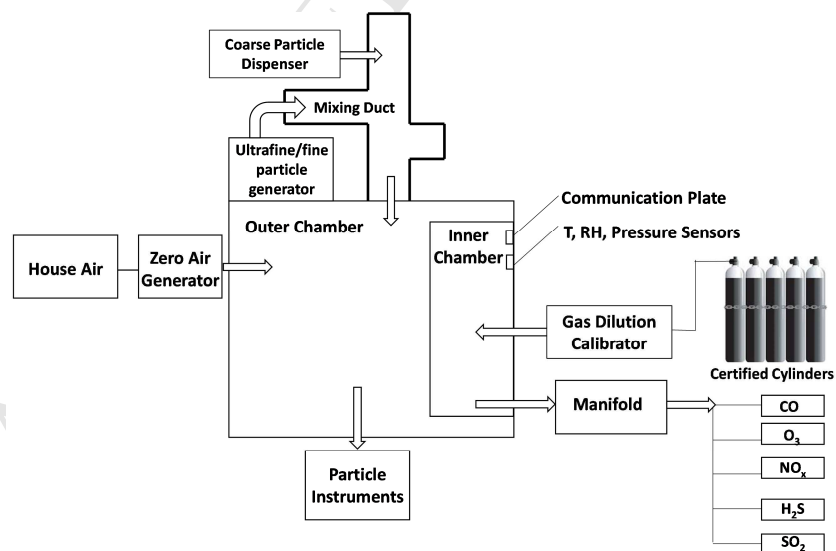


Figure 1. Schematic of chamber system

The chamber system consists of:

- i) a professional grade environmental test chamber (G-Series Elite, model GD-32-3-AC, Russells, Holland, MI) capable of creating a wide range of temperature and relative humidity. This includes a stainless steel rectangular-

shaped enclosure (1.3 m³, here referred to as “outer chamber”), a heating/cooling system for controlling the test temperature, a humidifier/de-humidifier for varying the relative humidity;

- ii) custom-made Teflon-coated stainless steel cylindrical-shaped enclosure (0.11 m³, here referred to as “inner chamber”) installed inside the “outer chamber” and used for gas sensors testing;
- iii) a dry, gas- and particle-free “zero-air” generation system (described in details in Supporting Information);
- iv) an aerosol generator (model AGK 2000, PALAS, Germany) to generate ultrafine/fine particles;
- v) a particle dispenser (model SAG 410/U, TOPAS, Germany) to dispense fine/coarse particles;
- vi) a dynamic dilution calibrator with an internal ozone generator (model T700U by Teledyne API, San Diego, CA);
- vii) an array of FRM, FEM and BAT reference instruments (Table 1), whose sampling inlets are in-line with the chamber and about 20 cm from the sensor trays (see Figure S-1. b);
- viii) an integrated computer software that controls the various operating/experimental parameters, T/RH set-points and reference instrument data logging.

Specifically, the integrated computer software (Polidori et al. 2016) allows for the design of extensive sensor testing experiments using programmed sequences, and consists of three main integrated components: a custom program sequence runner for setting chamber T/RH and for controlling PM concentrations in aerosol experiments (Figure S-2), a Teledyne APIcom software for controlling concentrations in gas experiments (Figure S-3), and a WinAQMS Data Acquisition and Control System (Figure S-4) (WinAQMS Air Quality Monitoring Software, Ecotech, RI). The software provides full control over the chamber system, enabling measurements to be recorded, automatic calibrations to be performed and system errors to be monitored. WinAQMS allows collection of data directly from the gas and particle instruments in digital format, thus eliminating digital to analog and analog to digital conversion errors.

Table 1. Reference instruments used in the laboratory low-cost sensor evaluation

Manufacturer	Model	Measures	Range	LDL	Measurement Technique	Method	Flow rate (LPM)
GRIMM	EC180	PM ₁₀ , PM _{2.5} , and PM ₁ environmental dust monitor	0.1-1500 µg/m ³	0.1 µg/m ³	light scattering	FEM	5
TSI	Model 3321	aerodynamic particle sizer, 0.5 to 20 µm aerodynamic sizing	0-10000 particles/m ³	0.001 particle/m ³	double-crest optical	BAT	0.5
TSI	Model 3091	fast mobility particle sizer, 5.6 to 560 nm	N/A	N/A	electrical mobility	BAT	5
Ecotech	Serinus 55	hydrogen sulfide	0-20 ppm	0.4 ppb	UV fluorescent radiation, external thermal converter	FRM	0.675
Ecotech	Serinus 10	ozone	0-20 ppm	0.5 ppb	nondispersive UV photometer	FEM	0.5
Ecotech	Serinus 40	nitric oxide, nitrogen dioxide, and nitrogen oxides	0-20 ppm	0.4 ppb	gas-phase chemiluminescence detection	FRM	0.6
Ecotech	EC 9850T	sulfur dioxide trace analyzer	0-20 ppm	0.3 ppb	fluorescence spectroscopy	FRM	0.75
Ecotech	EC 9830T	carbon monoxide trace analyzer	0-20 ppm	200 ppb	infrared photometric detection and gas filter correlation	FRM	1

2.2 Aerosol and gaseous test environments

Controlled aerosol environments are generated in the rectangular-shaped outer chamber to conduct PM sensor evaluations. As previously mentioned, the system is equipped with two particle generators, an ultrafine/fine particle generation system PALAS AGK 2000 and a fine/coarse particle dispenser TOPAS SAG 410/U. In the PALAS particle generator, test aerosol is generated in a consistent and systematic manner in terms of particle count/mass concentrations and particle size distribution. A specially developed nozzle prevents salt crystallization at the nozzle outlet. The liquid suspension is dried in a vertical dryer (model TR2000, PALAS, Germany) that has a smooth 90 degrees elbow shape. The dried aerosol is then diluted with dry “zero-air” into a mixing duct, and directed into the outer chamber. The particle concentration and size distribution can be adjusted by changing the salt type and concentration as well as by varying the aerosol injection frequency and duration in the chamber using the integrated custom software (see Section 2.1). The TOPAS dust dispenser is used to introduce fine/coarse particles in the outer chamber. Standard dry dust (e.g., Arizona test dust ISO 12103 A4 Coarse Grade or dolomite powder) is dispensed into the chamber through a two-step process: first, dust is continuously supplied from a dust reservoir onto the rotating belt, and subsequently, the dust material is injected into the mixing duct and then the outer chamber with compressed “zero-air”. The first step is key to achieve a stable and reproducible particulate concentration, but is quite challenging, especially when the target concentration is very low. Key dispenser components such as the toothed belt and the scraper ensure the delivery of consistent amounts of dust.

In a typical fine particle ($PM_{2.5}$) experiment testing a PM sensor, the outer chamber is first conditioned to a target temperature and RH. Aqueous potassium chloride solution (KCl, 17% by weight) is nebulized in the aerosol generator and is subsequently dried in the PALAS dryer. Dry particles are then directed into the chamber. The chamber operates in a dynamic mode at a flowrate of about 23 L/min. On average, 2 to 3 hours are needed to achieve a stable particle concentration and size distribution in the chamber, which is between 2 to 3 chamber mean residence times for a chamber volume of 1.3 m^3 .

Controlled gaseous atmospheres are generated in the cylindrical-shaped inner chamber where gas sensors are installed for testing. The relatively low inner chamber volume of 0.11 m^3 (about $1/12^{\text{th}}$ of the outer chamber) allows to decrease the time needed to reach a stable gas concentration in the chamber and, consequently, to reduce the consumption of certified standard gases used for experiments. The inner chamber is coated with polytetrafluoroethylene (PTFE) to provide an inert surface for reactive gases (e.g., O_3 and H_2S).

In a typical experiment of testing a gas sensor, the inner chamber is first conditioned to the target temperature and RH. The targeted gas pollutant(s) from a certified compressed gas cylinder (or in the case of ozone, generated by the dilution calibrator) is first passed through a gas dilution calibrator where the appropriate test flow is adjusted and mixed with pre-conditioned zero-air inside the inner chamber. The dynamic dilution calibrator is capable of producing a broad range of gas pollutant concentrations.

2.3 Chamber preparation and sensor evaluation

Prior to a laboratory testing experiment, the chamber wall surfaces are wiped off with isopropyl alcohol and the sensors to be tested are mounted on the supporting Teflon-made trays (Figure S-1 b). All sensors are pre-treated with routine set-up procedures, such as filter replacement (when necessary), zero calibration, flow rate check, date/time synchronization, and battery charge. Power and software control cables are connected through the communication plate located on the right-hand side of the outer chamber. The chamber door is then closed and dry, particle- and gas-free air is flushed through the system to provide a clean background for a new experiment.

The general evaluation procedure is based on a side-by-side comparison between the sensor(s) to be tested and the reference FRM, FEM and BAT instrument(s) measuring the same pollutant(s). During a typical experiment, the concentration of the target

pollutant is increased from zero-air baseline to the first concentration level. The target pollutant concentration is then kept stable for a set period of time at the steady-state stage. This procedure is repeated multiple times over multiple pollutant concentrations spanning from very low (e.g., below typical ambient conditions or less than 50% of the U.S EPA National Ambient Air Quality Standards (NAAQS)) to very high (e.g., well above ambient conditions or more than 200% of the U.S. EPA NAAQS), and under
 5 different temperature (e.g., 5 – 35 °C) and RH levels (e.g., 20 – 80%). Performance-related parameters, similar to those established by EPA, JRC and other organizations that have conducted similar sensor testing experiments (Spinelle et al. 2013; Spinelle et al. 2014; Williams et al. 2014; Williams et al. 2015) are quantitatively defined and include intra-model variability, accuracy, precision, lower detection limit, linear correlation coefficient (R^2), effect of interferent species, climate susceptibility, and data recovery (Polidori et al. 2016). Most of these parameters, except for the linear correlation coefficient, are evaluated
 10 based on data acquired from the steady-state stages.

3 Results and Discussion

3.1 Chamber temperature and relative humidity control

Studies have shown that temperature and relative humidity can have a negative impact on sensor performance and, consequently, affect the reported pollutant concentration. Commercially available metal oxide gas sensors are mainly made of tin dioxide
 15 (SnO_2) deposited on porous alumina or silica. At high humidity level, adsorption of water vapor and potential moisture condensation decrease the SnO_2 baseline resistance, resulting into reduced sensitivity. The effective surface area available for sensing the target pollutant also drops in time due to moisture adsorption (Sohn et al. 2008). In addition, prolonged exposure to high humidity levels leads to the formation of stable chemisorbed hydroxide ions (OH^-), causing sensitivity degradation or drifted baseline values (Wang et al. 2010). For optical particle sensors, relative humidity is known to affect their performance
 20 due to the condensational growth of the particles' hygroscopic components. The enlargement of the particle size contributes to the overestimation of the particle mass concentration (Chakrabarti et al. 2004; Fischer and Koshland 2007).

Temperature, on the other hand, does not generally influence sensor performance as much as relative humidity. However, in some particle sensors, the air sample is conveyed by a current generated using a built-in heater. At extreme temperatures, the flow rate of the current could be altered, and consequently affect the resulting laser or infrared particle count readings (Wang et
 25 al. 2015). Manufacturers use proprietary algorithms to convert particle count into particle mass concentration. In some cases, the algorithm takes into account not only the particle density and shape, but also the operating temperature and humidity.

For the above reasons, it is imperative that the sensors are evaluated under representative environmental conditions. As suggested by the specification, our chamber system is capable of reaching a broad range of temperature and relative humidity conditions (T range: -32 °C to +177 °C; RH range: 10% to 95%). The chamber's capability to create stable and reproducible
 30 temperature and relative humidity conditions was demonstrated in two separate experiments. In the first experiment, the temperature set point was increased by 5 °C every 30 minutes from 5 to 35 °C, while maintaining relative humidity constant at 40%. As shown in Figure 2 (a), the chamber was capable of delivering accurate temperature control. At each step, the temperature reached the target value within three minutes from the previous set point and remained stable throughout the 30 minutes period. Previous studies have also demonstrated that testing chambers are capable of controlling temperature conditions
 35 within ± 2 °C (Zhao et al. 2015).

Regarding humidity control, two different chamber dehumidification processes are available and may be used. If the temperature and humidity combination falls in the shaded area in Figure S-2 T-RH performance envelop, dehumidification is accomplished by eliminating excess of water at the cooling coils. For conditions that lie outside the shaded area in T-RH performance envelope

(Figure S-2), the dehumidification of the chamber can be achieved by purging the system with dry zero air. In an experiment to validate chamber's humidity control, the chamber was held at a constant temperature of 20 °C, and RH was ramped from 20% to 80% at 10% intervals. Each RH level was maintained for 60 minutes. In Figure 2 (b), at the lowest set point of 20%, RH fluctuated between 20% and 23%. This was somewhat expected because this T-RH combination is outside the shaded T-RH performance envelop, and therefore RH could not be maintained at such low level without purging dry zero-air. At higher RH set points, the measured RH fluctuations around the target values were between $\pm 0.8\%$ and $\pm 3.9\%$ and were considered very small and within the acceptable range. It should be noted that in previous chamber studies, RH conditions were not accounted for (Austin et al. 2015; Sousan et al. 2016a; Zhao et al. 2015). In other cases where water bubblers were used to generate certain RH levels, gas flows were split based on calculations, and only the theoretical RH level was reported. Actual RH variations in the chamber were not monitored nor discussed (Wang et al. 2015; Wei et al. 2013). Here, the two experiments clearly demonstrated the chamber's capability to produce a broad range of environmental conditions that can be used to appropriately test sensors' climate susceptibility.

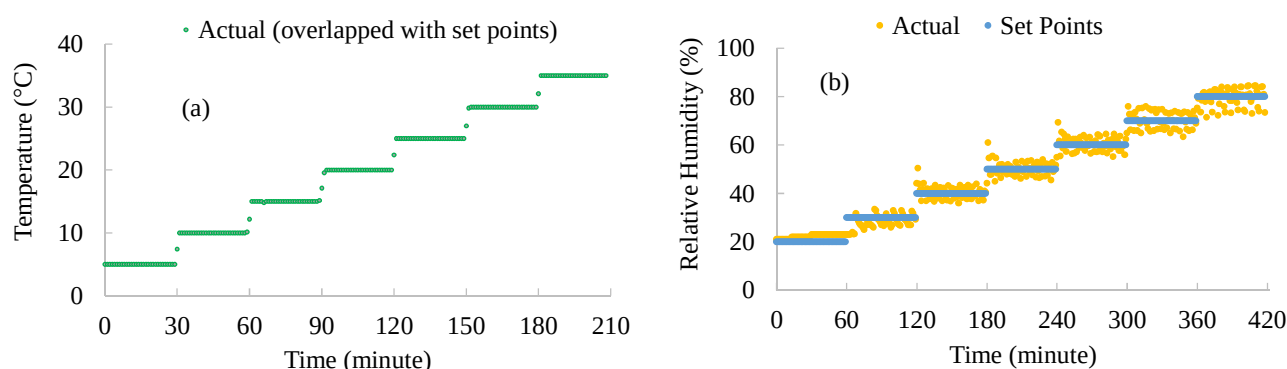


Figure 2. Chamber's response with (a) 5 °C increment temperature ramping (at constant 40% RH); (b) 10% increment RH ramping (at constant 20 °C)

3.2 Generation of stable testing atmospheres

The chamber's ability to generate stable aerosol and gas atmospheres is crucial for performing sensor evaluation, as many testing parameters are calculated based on the data acquired under steady-state conditions. Thus, a set of experiments was conducted to fully evaluate the chamber's ability to produce stable concentrations of laboratory generated particles, CO (an unreactive gas), and O₃ (a reactive gas). Stability for each pollutant was evaluated at five indicative concentration levels from "very low" to "very high". The "very low" level is chosen to represent a pollutant concentration below the U.S. EPA NAAQS requirement. It is also close to the lowest concentration the chamber system could generate or the reference analyzers could detect. The "low" and "medium" pollutant levels are similar to the average ambient pollutant concentration level and 50% above the average ambient pollutant concentration level, respectively. The "high" pollutant level is at least 100% higher than the average NAAQS standard level. The "very high" pollutant level is close to the maximum concentration the chamber could generate or the reference analyzers could accurately detect. Table S-1 shows some indicative concentration set-points of different pollutant types.

3.2.1 Stability of aerosol experiments

During an aerosol atmosphere experiment, the chamber was first allowed to reach the target concentration (in about 2 hours), then the experiment was continued for another 3 hours to evaluate chamber's ability to maintain the stable concentration. Stability is evaluated by estimating the relative percentage of standard deviation (% RSD) based on the measured concentration

variations at each steady-state. All experiments were carried at 20 °C and 40%. As shown in Figure 3 (a), five target aerosol concentrations, from “very low” to “very high” were generated by the ultrafine/fine particle generator. The steady-state concentrations were reached within two hours, and were measured to be 6.4, 25.7, 36.4, 125.7, and 229.7 $\mu\text{g}/\text{m}^3$. Stability, expressed by one SD of the concentration variations during the steady-state, was found to be 0.2, 0.5, 0.5, 3.2 and 4.3 $\mu\text{g}/\text{m}^3$ for the five concentration mentioned above, respectively. Larger variations were observed at higher pollutant concentrations. The % relative SDs were between 1.4% and 3.1% over the entire duration of steady-state for each experiment (last 3 hours), suggesting that the chamber is capable of maintaining excellent stability over a wide concentration range. To achieve steady-state conditions, the rate of aerosol injection in the chamber must equal the rate of the amount of aerosol removed from the chamber. The observed stability is achieved mainly due to i) the particle generator’s ability to provide precise and consistent amount of aerosol during each injection ii) efficient particle mixing inside the chamber using a set of two fans installed on the back wall of the outer chamber that ensure a homogenous and uniform environment. The FEM GRIMM reference dust monitor sampled at a constant flowrate of 5 L/min.

Particle decay and settling was also investigated. An aerosol decay experiment, described in the Supporting Information, showed that the particle decay process in the chamber tracked well with the first order kinetics, which was also consistent with previous study (Cocker et al. 2001). The modelled decay process has a reaction rate constant of 0.004 min^{-1} and a half-life of 173 minutes (shown in Figure S-3). In addition, the mixing fan speed had an effect on the ultimate particle concentrations, mainly on the portion lost onto the chamber wall. Intuitively, the higher the fan speed, the higher the chance of a particle hitting the chamber walls, and the lower the eventual test aerosol mass concentration. Figure S-4 shows the distinct decrease in aerosol concentration as the mixing fan frequency increased from 20 to 40 Hz, and from 40 to 60 Hz resulting into a higher speed and a distortion in the uniform mixing of particles inside the chamber. Thus, the mixing fan speed was kept at a constant frequency of 25 Hz in all experiments.

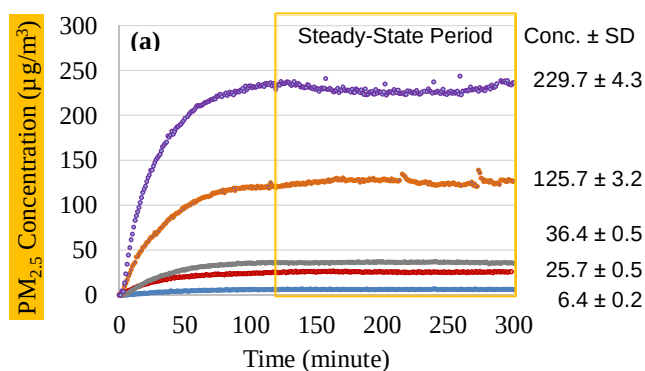


Figure 3a. Stability of various particle concentration levels.

3.2.2 Stability of gas experiments

To evaluate the stability of the gas atmospheres, experiments with CO and O₃ were conducted in the inner chamber. The chamber was allowed to ramp up to the target concentration (in about 30 minutes), then the experiment was continued for another 60 minutes. For CO, the steady-state concentrations were reached in about 30 minutes. At the “very low” target concentration of 1 ppm, the variation at steady-state was negligible. At the highest target concentration of 22.2 ppm, the variation was 0.5 ppm, or 2.3% RSD. For O₃, despite of its reactive nature, the system maintained stable O₃ concentrations at all levels, ranging from 45 to 301 ppb. The variation at the 45 ppb target concentration was only 1 ppb, while the variation at the 301 ppb

target concentration was only 8 ppb, which represented 2.2% and 2.7% RSD, respectively. Previous calibration study of CO, NO, and NO₂ sensors in an acrylic glass chamber also showed relatively stable gas concentrations over a one hour period (Mead et al. 2013). Similar to the aerosol atmosphere, the gas atmosphere steady-state condition in the chamber was achieved by balancing the input and output of the pollutant gas. Certified gas and “zero-air” dilution flow were regulated by a mass flow controller. As shown in Figures 3 (b) and (c), regardless of the reactivity of the gas pollutant, the chamber system was able to maintain a stable gas concentration needed for sensor evaluation.

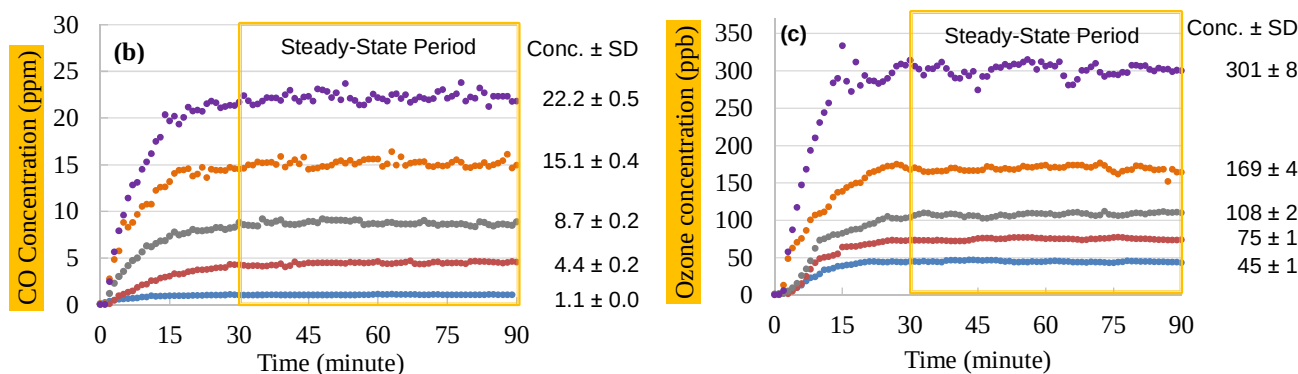


Figure 3b-c. Stability of various gaseous concentration levels.

3.3 Reproducibility of stable aerosol and gaseous atmospheres

As discussed in the previous sections, the temperature and relative humidity can be accurately controlled over a wide range (Section 3.1). The chamber is capable of maintaining stable aerosol and gaseous atmospheres (Section 3.2). To compare different types of sensors, each characteristic parameter needs to be evaluated under similar testing atmospheres. The chamber system is expected to generate similar pollutant concentrations in repeated experiments. Therefore, following stability tests, the reproducibility of chamber performance was demonstrated in several experiments.

3.3.1 Reproducibility of aerosol experiments

For aerosol atmosphere, the reproducibility of experiments were evaluated for low, medium, and high PM concentrations (relative to U.S. EPA NAAQS). For each concentration, three repeated runs were used to determine the reproducibility of the chamber system. As shown in Figure 4 (a), the average pollutant concentrations from three repeated experiments were plotted. At the steady-state stage, error bars (one SD of three concentrations reported in the repeated experiments) were used to represent the variation in pollutant concentrations among the three experiments. At low PM level, the steady-state concentrations for three repeated runs were 15.9, 15.8, and 19.6 $\mu\text{g}/\text{m}^3$, respectively. It represented 12.6% variation (calculated using one SD divided by average concentration of three repeated runs) of the average value at low concentration. At medium PM level, the steady-state concentrations for the runs were 36.4, 41.4, and 44.6 $\mu\text{g}/\text{m}^3$, which represented 10.0% variation. At high PM level, the steady-state concentrations for the runs were 229.2, 219.3, and 233.6 $\mu\text{g}/\text{m}^3$, and the variation was only 3.2%. Considering the large volume of the chamber, the system demonstrated the ability of generating highly reproducible PM concentrations.

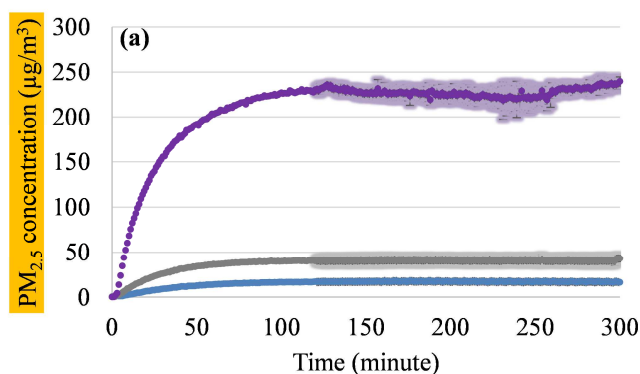


Figure 4a. Reproducibility of various particle concentration levels.

3.3.2 Reproducibility of gas experiments

To evaluate the chamber's ability to generate reproducible gaseous atmospheres, experiments in the inner chamber were run using an unreactive gas CO and a reactive gas O₃. Similar to the aerosol experiments, representative concentrations at low, medium, and high pollutant levels were tested.

The CO concentration reproducibility was tested at 1, 8, and 22 ppm, showed in Figure 4 (a). For each concentration, the variation between three runs was under 7.5%. The O₃ concentrations reproducibility was tested at 45, 120, and 300 ppb, showed in Figure 4 (b). For each concentration, the variation between three runs was under 7.0%.

In summary, our results indicate that the chamber system has excellent reproducibility in creating similar conditions. It is suitable for evaluating air quality sensors of different types, as they can be compared against each other.

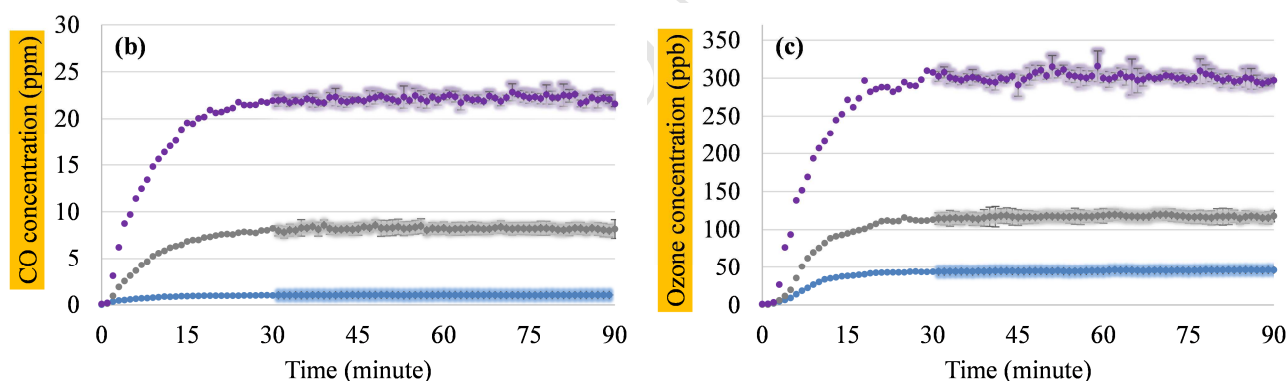


Figure 4b-c. Reproducibility of various gaseous concentration levels.

3.4 Laboratory sensor evaluation and testing design

The laboratory testing procedures are described in AQ-SPEC's Laboratory Evaluation of Low-Cost Air Quality Sensors – Laboratory Setup and Testing Protocol (Polidori et al. 2016). In general, the laboratory testing consists of two phases: (I) concentration ramping at average ambient conditions; (II) sensor climate susceptibility evaluation. In Phase I, the chamber is first conditioned to 20 °C and 40% RH, and a concentration ramping experiment begins. A total of 5 concentration steps are selected to simulate a diverse pollutant concentration profile from “very low” to “very high” (relative to the U.S. EPA NAAQS). Every concentration step change occurs only after a stable pollutant concentration has been reached in the chamber and a sufficient time is allowed in the steady-state period. Each step is held for 150 minutes during an aerosol test and 40 minutes during a gas test.

Sample experiments are presented in Figure 5. Data is analyzed to quantitatively evaluate sensor's accuracy, lower detection limit, and linear correlation coefficient. In Phase II, pollutant concentration, temperature and RH are each varied at three different levels (low, medium, and high) for a total of 27 different combinations. Specifically, low, medium, and high pollutant concentrations are selected based on Table S-1. Temperature is varied between low (5 °C), medium (20 °C), and high (35 °C).
 5 RH is varied between low (15%), medium (40%), and high (65%). A complete list of the conditions used is listed in the Supplemental Information. Data acquired from this phase are used to derive information on sensor's climate susceptibility and precision. For gaseous sensors, the effect of interferences is also examined. Depending on the sensor's target pollutant type and its detecting technology, the interferences and their concentrations are chosen as recommended by 40 CFR Part 53 Table B-320 or from information provided in other studies (Austin et al. 2006; Mead et al. 2013). For example, for an electrochemical ozone
 10 sensor, NO₂ and sulfur dioxide (SO₂) are considered interferences, and sensor's response towards either of these compounds is investigated separately.

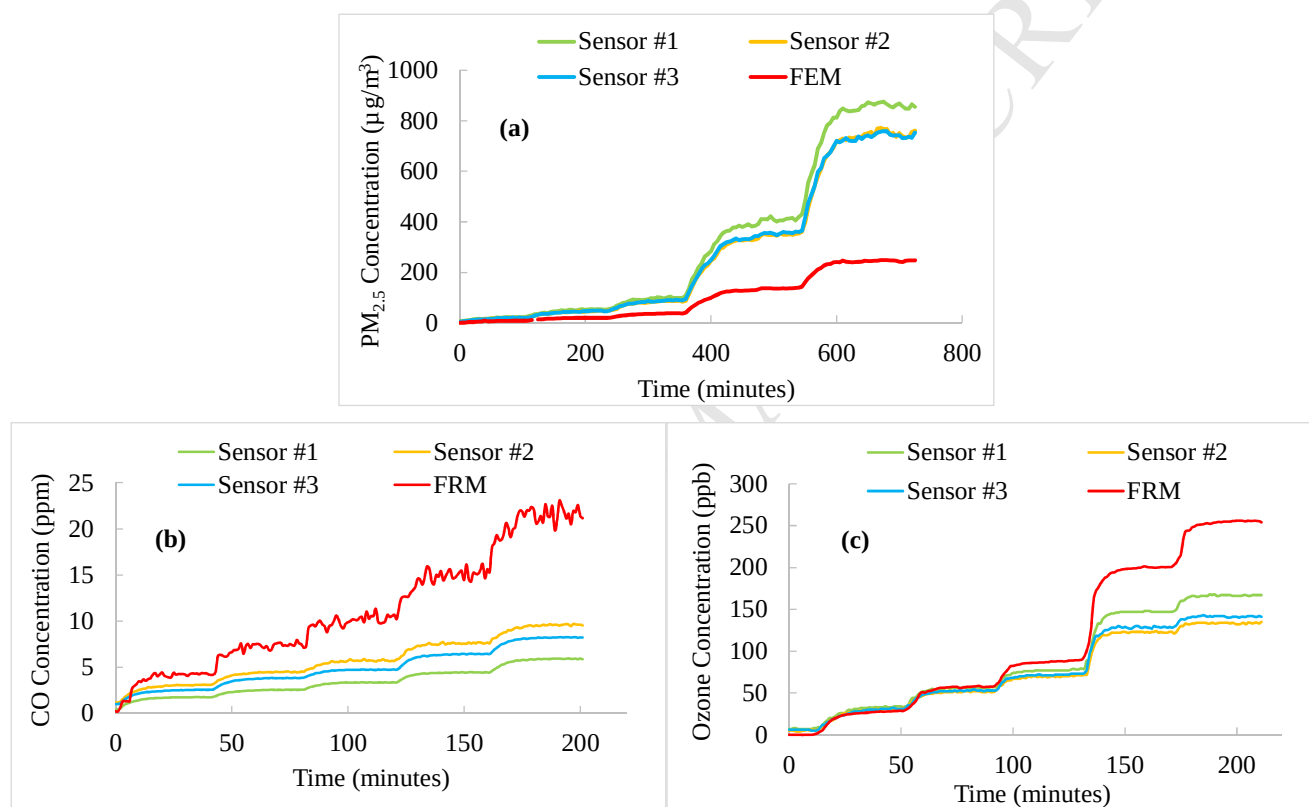


Figure 5. Examples of sensor evaluation experiments: (a) aerosol, (b) CO, and (c) O₃ atmospheres

4 Conclusion

This state-of-the-art laboratory chamber is the first of its kind designed and developed to systematically evaluate the performance of commercially available air quality sensors. A series of experiments were conducted to demonstrate the system's capability of generating stable and reproducible aerosol and gaseous atmospheres, as well as its capability of simulating a wide range of temperature (T) and relative humidity (RH) conditions. An environmental chamber system of this type is capable of engaging in
 20 comprehensive characterizations, validations and calibrations of air quality sensors. Preliminary testing results have shown important information regarding sensor accuracy, precision, detection limit, correlation coefficient, interferences, and climate

susceptibility. Information from both the field and laboratory testing should be taken into consideration to better understand sensor performance under baseline, ambient and extreme T/RH conditions and pollutant levels.

Institutions and organizations interested in evaluating the performance of low cost air quality sensors should work together to develop standardized field and laboratory testing procedures. This paper provides other researchers with a well proven method to fully characterize the performance of different sensor types. Low-cost sensors can already be used for augmenting ambient air monitoring networks, community based monitoring, screening for hot spots, science education, and qualitative personal monitoring. Thus, it is important for the scientific community to fully understand the capabilities and limitations of this emerging technology.

Disclaimer

- Mention of trade names or commercial products does not constitute endorsement or recommendation for use. Potential consumers are welcome to review the completed field and laboratory evaluation reports at www.aqmd.gov/aq-spec.

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